

Sharp flat-spectral volume asymptotics and typical spectra of absolutely PPT states

Yongxian Zhang

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School of Computer Science and Engineering, South China University of Technology, Guangzhou, China. Correspondence: mxymmxym1@gmail.com. ORCID: 0009-0000-3864-3536.

Abstract. We determine the sharp asymptotic flat-spectral volume of absolutely positive-partial-transpose (APPT) states for every fixed smaller local dimension, including the leading coefficient and a relative error of order inverse total dimension. A full-rank Schmidt test defines an explicit containing affine simplex; a uniform Schur-complement estimate and Dirichlet concentration show that its non-APPT fraction vanishes at that rate. We also obtain a common conditional spectral limit shape for APPT and absolutely separable states, a truncated exponential limiting eigenvalue density, typical purity, and an exact asymptotic comparison with the known inner polytope. The separability consequence uses an established spectral-ratio criterion. These are analytic proofs, not Lean formalizations. The unrestricted maximal-purity conjecture and equality of APPT and absolute separability remain separate unresolved problems.

Written analytic proof; not Lean-formalized. The measure is flat Lebesgue measure on the eigenvalue probability simplex, not Hilbert–Schmidt or Bures measure on density matrices. The smaller local dimension is fixed. This note does not solve the unrestricted maximal-purity conjecture or prove APPT equals absolute separability.

1. Statements and scope

Fix an integer $m \geq 2$, let $n \geq m$ and $D = mn$, and set

$$R = \frac{m(m-1)}{2}, \quad S = \frac{m(m+1)}{2}, \quad \alpha = \frac{m-1}{2}, \quad \beta = \frac{m+1}{2} = \alpha + 1.$$

A density matrix is APPT if partial transpose remains positive after every global unitary conjugation. It is absolutely separable (AS) if every conjugate is separable. Define V_E as the volume of the corresponding spectra divided by the volume of the probability simplex. Permutation invariance gives the same ratio on its nonincreasing ordered section. There is no Vandermonde weight.

Put

$$q_m = \frac{\alpha^\alpha}{\beta^\beta}, \quad C_m = \frac{\sqrt{2\pi} m^{m^2-1} \alpha^{R+1/2} \beta^{S-1/2}}{R!(S-1)!}.$$

Theorem 1. For each fixed $m \geq 2$, as n tends to infinity,

$$\boxed{V_{\text{APPT}}(m, n) = C_m \sqrt{D} q_m^D (1 + O_m(D^{-1}))}. \quad (1)$$

In particular,

$$V_{\text{APPT}}(2, n) \sim \frac{9\sqrt{3\pi}}{2} \sqrt{n} (4/27)^n, \quad V_{\text{APPT}}(3, n) \sim \frac{2916\sqrt{3\pi}}{5} \sqrt{n} 64^{-n}. \quad (2)$$

An explicit finite-dimensional sandwich underlies (1). Define

$$W_{m,D} = \frac{m^{m^2-1} \Gamma(D) \Gamma(\alpha D + R + 1)}{(D+m)^R (D-m)^{S-1} R! (S-1)! \Gamma(\beta D - S + 1)}. \quad (3)$$

For $K = m^2 - 2$ and $c = 2m(m+1)$, put

$$E_{m,D} = \frac{576m^2}{D+1} + \frac{144c^2 K(K+1)}{D(D+1)} + \frac{6c^2 K(K+1)}{D}. \quad (4)$$

For $D \geq 12m(S-1)$, as well as $D = mn$ and $n \geq m$,

$$(1 - E_{m,D}) W_{m,D} \leq V_{\text{APPT}}(m, n) \leq W_{m,D}. \quad (5)$$

The lower bound may be negative for moderate D ; its constants are not optimized.

Theorem 2. Using [KBNKB, Supplemental Lemma 2],

$$V_{\text{AS}}(m, n) = \Theta_m(\sqrt{D} q_m^D). \quad (6)$$

For either APPT or AS, sample the spectrum uniformly under this flat measure and order its entries. Let

$$L = \log(\beta/\alpha), \quad a = \alpha L, \quad b = \beta L, \quad Q_m(s) = a + \log \frac{\beta}{\alpha + s} \quad (0 \leq s \leq 1).$$

Then

$$\boxed{\max_{1 \leq i \leq D} |D\lambda_i - Q_m(i/D)| \longrightarrow 0 \quad \text{in probability.}} \quad (7)$$

Thus the scaled empirical eigenvalues have limiting density

$$f_m(x) = \beta e^{a-x} \mathbf{1}_{[a,b]}(x), \quad (8)$$

and typical scaled purity satisfies

$$D \text{Tr}(\rho^2) \longrightarrow 2 - \alpha\beta[\log(\beta/\alpha)]^2. \quad (9)$$

The scaled minimum and maximum converge to a and b , respectively, and the extreme-eigenvalue ratio tends to $(m+1)/(m-1)$. These are typical-spectrum conclusions, not maximum-purity bounds.

Corollary 3. For the inner polytope of [AKW, Proposition 6.14],

$$\frac{V_{\text{APPT}}}{V_{\mathcal{P}}} \longrightarrow \frac{m^{m^2-1} \alpha^{R-m+1} \beta^{S-m} (m-1)! m!}{4^{m-1} R! (S-1)!}. \quad (10)$$

The limit is 3 for $m = 2$ and $2187/40$ for $m = 3$.

Attribution. Hildebrand's spectral method and the permutation-invariant APPT formulation of [AKW] are established inputs. Section 6.3 of [AKW] gives the exact inner-polytope volume and numerical comparisons for $m = 2, 3, 4$; its stated theorem gives a lower bound, with a constant-factor comparison discussed numerically. [Tran] proves qubit volume order $\Theta(\sqrt{n}(4/27)^n)$ using explicit inner and outer bounds. Here the new argument is asymptotically full volume in an explicit containing affine simplex, its general leading coefficient, and the conditional limit shape. The AS criterion is credited to [KBNKB]. No exhaustive historical-priority or peer-review claim is made.

2. Physical tests and a uniform Schur bound

For a unit vector with Schmidt coefficients $s_i \geq 0$ and $\sum_i s_i^2 = 1$, the partially transposed rank-one projector has eigenvalues

$$s_i^2 \ (1 \leq i \leq m), \quad \pm s_i s_j \ (i < j), \quad D - m^2 \text{ zeros}. \quad (11)$$

This follows by transposing matrix units; the off-diagonal two-dimensional blocks have symmetric and antisymmetric eigenvectors. Partial transpose is self-adjoint for the trace pairing:

$$\langle \psi, (U \rho U^*)^\Gamma \psi \rangle = \text{Tr}(U \rho U^* (|\psi\rangle\langle\psi|)^\Gamma). \quad (12)$$

The minimum over the unitary orbit is the oppositely ordered eigenvalue pairing: squared absolute unitary entries form a doubly stochastic matrix, rearrangement gives the bound, and a permutation unitary attains it.

For sorted nonnegative λ define

$$h(\lambda) = \sum_{j=D-S+1}^D \lambda_j - \sum_{j=1}^R \lambda_j. \quad (13)$$

Taking all Schmidt coefficients equal in (11) proves APPT implies $h \geq 0$.

Assign the largest R eigenvalues to slots a_{ij} for $i < j$, and the smallest S to slots b_{ij} for $i \leq j$, in arbitrary orders. Form the real symmetric matrix

$$(L_{a,b})_{ii} = 2b_{ii}, \quad (L_{a,b})_{ij} = b_{ij} - a_{ij} \quad (i < j). \quad (14)$$

If all these matrices are PSD, λ is APPT. Indeed, for strictly positive Schmidt coefficients the worst trace pairing puts the top R entries on negative terms and the bottom S on positive terms, with some slot assignments. Its value is $s^T L_{a,b} s / 2$, while middle eigenvalues pair with zeros. Lower Schmidt ranks follow by continuity. Local Schmidt basis changes are absorbed into global unitaries. No numerical chamber enumeration is needed.

Write

$$\bar{a} = R^{-1} \sum_{j=1}^R \lambda_j, \quad \bar{b} = S^{-1} \sum_{j=D-S+1}^D \lambda_j, \quad g = \bar{a} + \bar{b},$$

$$A = \lambda_1 - \lambda_R, \quad B = \lambda_{D-S+1} - \lambda_D, \quad \epsilon = (m-1)A + (m+1)B.$$

For $m = 2$, $A = 0$. Let $e = (1, \dots, 1) / \sqrt{m}$, J be the all-ones matrix, and $L_0 = gI + (\bar{b} - \bar{a})J$. Every assignment satisfies

$$e^T L_0 e = 2h/m, \quad L_0|_{e^\perp} = gI, \quad e^T (L_{a,b} - L_0) e = 0. \quad (15)$$

The last equality is exact because the sums of the a- and b-slots do not change. Perturbation entries are bounded by $2B$ on the diagonal and $A + B$ off it, so the symmetric operator norm is bounded by the maximum absolute row sum ϵ . The $e^\perp$ block is therefore at least $(g - \epsilon)I$, the coupling to e has norm at most ϵ , and the e, e entry remains exactly $2h/m$. Completing the square gives the sufficient condition

$$\boxed{\lambda_D \geq 0, \quad g > \epsilon, \quad (2h/m)(g - \epsilon) \geq \epsilon^2 \implies \lambda \text{ is APPT.}} \quad (16)$$

The same estimate controls every assignment at once.

3. The containing affine simplex

The ordered probability simplex has vertices $u_k = (1/k, \dots, 1/k, 0, \dots, 0)$. Its barycentric coordinates are

$$x_k = k(\lambda_k - \lambda_{k+1}) \quad (k < D), \quad x_D = D\lambda_D, \quad \sum x_k = 1. \quad (17)$$

Temporarily retain $x_k \geq 0$ for $k < D$ while allowing $x_D < 0$. This auxiliary affine region is not a space of density matrices. Direct counting in (13) gives

$$h(u_k) = \begin{cases} -1 & k \leq R, \\ -R/k & R < k \leq D - S, \\ (k - D + m)/k & D - S < k \leq D. \end{cases} \quad (18)$$

The middle range may be empty. The maximum is uniquely $h(u_D) = m/D$. Set $c_k = \frac{m/D}{m/D - h(u_k)}$ for $k < D$. Explicitly,

$$c_k = \begin{cases} m/(D+m) & k \leq R, \\ k/(k+\alpha D) & R < k \leq D-S, \\ m(D-l)/[l(D-m)] & k = D-l, 1 \leq l < S. \end{cases} \quad (19)$$

These numbers are positive. The simplex

$$T = \{x_k \geq 0 \ (k < D), \ \sum_{k < D} x_k/c_k \leq 1\}, \quad x_D = 1 - \sum_{k < D} x_k \quad (20)$$

contains all ordered APPT spectra. Its intersection with $x_D \geq 0$ is exactly the physical ordered linear relaxation $h \geq 0$.

Put $x_k = c_k y_k$ for $k < D$ and $y_D = 1 - \sum_{k < D} y_k$. Then y ranges over the standard probability simplex, with constant affine Jacobian, and

$$h(\lambda) = (m/D)y_D. \quad (21)$$

The volume of T divided by that of the ordered probability simplex is $\prod_{k < D} c_k$. Indeed the affine map from barycentric coordinates to λ has the same Jacobian in numerator and denominator. Multiplying (19),

$$\frac{\text{vol}(T)}{\text{vol}(\Delta_D^\downarrow)} = \left(\frac{m}{D+m}\right)^R \prod_{k=R+1}^{D-S} \frac{k}{k+\alpha D} \prod_{l=1}^{S-1} \frac{m(D-l)}{l(D-m)} = W_{m,D}. \quad (22)$$

Cancellation gives precisely (3), including an empty middle product. All Gamma arguments are integers in product dimensions since $\alpha D = m(m-1)n/2$ is integral.

4. Almost-full physical APPT volume inside T

Under uniform volume on T , y is Dirichlet(1, ..., 1) on D entries. Elementary simplex integration gives

$$\mathbb{E}y_i = 1/D, \quad \mathbb{E}y_i^2 = 2/[D(D+1)], \quad \mathbb{E}y_i y_j = 1/[D(D+1)] \quad (i \neq j), \quad (23)$$

and

$$\text{Var}(\sum_i d_i y_i) = \frac{D \sum_i d_i^2 - (\sum_i d_i)^2}{D^2(D+1)}. \quad (24)$$

Also $t = y_D$ has Beta(1, $D-1$) distribution independently of $(y_1, \dots, y_{D-1})/(1-t)$, which is uniform Dirichlet on $D-1$ entries. This follows by the change-of-variables Jacobian $(1-t)^{D-2}$, not an asymptotic independence assumption.

Assume $D \geq 12m(S-1)$, hence $D \geq 2m$. All $c_k \leq 2m$, and for $k \leq D-S$ one has $c_k \leq 2/(m+1) \leq 2/3$. Thus

$$\mathbb{E}x_D = 1 - D^{-1} \sum_{k < D} c_k \geq 1/3 - 2m(S-1)/D \geq 1/6.$$

Taking $d_k = c_k$ for $k < D$ and $d_D = 0$ in (24), Chebyshev gives

$$\Pr_T(x_D < 1/12) \leq 576m^2/(D+1). \quad (25)$$

Let $J = \{1, \dots, R-1\} \cup \{D-S+1, \dots, D-1\}$. It has $K = m^2 - 2$ entries, none equal to D , and its ranges are disjoint. Put $Z = \sum_{j \in J} y_j$. The endpoint gaps are exactly

$$A = \frac{m}{D+m} \sum_{j=1}^{R-1} y_j/j, \quad B = \frac{m}{D-m} \sum_{l=1}^{S-1} y_{D-l}/l. \quad (26)$$

It follows that $\epsilon \leq cZ/D$, $c = 2m(m+1)$. On the three simultaneous events

$$x_D \geq 1/12, \quad Z \leq 1/(12c), \quad y_D \geq 6c^2Z^2, \quad (27)$$

we have $\lambda_D \geq 1/(12D)$, $g \geq 1/(6D)$, $\epsilon \leq 1/(12D)$, and therefore

$$(2h/m)(g - \epsilon) \geq (2y_D/D)/(12D) \geq c^2Z^2/D^2 \geq \epsilon^2.$$

Thus (16) proves physical APPT on their intersection.

The second moment $\mathbb{E}Z^2 = K(K+1)/[D(D+1)]$ yields

$$\Pr_T(Z > 1/(12c)) \leq 144c^2K(K+1)/[D(D+1)]. \quad (28)$$

For the last event write $t = y_D$ and $Z = (1-t)Z'$. Then Z' is independent of t , with second moment $K(K+1)/[(D-1)D]$. The event $t < 6c^2(1-t)^2(Z')^2$ implies $t < 6c^2(Z')^2$. The beta CDF satisfies $\Pr(t < u) \leq \min(1, (D-1)u)$ for $u \geq 0$, by $1 - (1-u)^{D-1} \leq (D-1)u$ on $[0, 1]$. Consequently

$$\Pr_T(y_D < 6c^2Z^2) \leq 6c^2K(K+1)/D. \quad (29)$$

There is no incorrect independence assumption between Z and y_D . Applying the union bound to (25), (28), (29) proves

$$1 - E_{m,D} \leq \Pr_T(\lambda \text{ is APPT}) \leq 1.$$

Multiplication by (22) gives (5). In particular the entire affine simplex and its physical linear relaxation both have volume asymptotic to APPT, although the relaxation is not a finite-dimensional equivalence criterion.

5. Leading constant and inner-polytope comparison

For fixed $a > 0$ and fixed b , Stirling's formula gives

$$\Gamma(aD + b) = \sqrt{2\pi}(aD)^{aD+b-1/2}e^{-aD}(1 + O_{a,b}(D^{-1})).$$

Apply it to the three Gamma factors in (3). The exponentials cancel because $1 + \alpha = \beta$. Their combined D -power is $R + S - 1/2 = m^2 - 1/2$; the two other powers in the denominator contribute D^{m^2-1} . The remaining exponential is q_m^D and the coefficient is exactly C_m . This proves $W = C_m\sqrt{D}q_m^D(1 + O_m(1/D))$; (5) then proves Theorem 1.

The exact inner-polytope formula [AKW, Eq. (64)] similarly gives

$$V_{\mathcal{P}} = C_m^{\text{in}}\sqrt{D}q_m^D(1 + O_m(D^{-1})), \quad C_m^{\text{in}} = \frac{4^{m-1}\sqrt{2\pi}(\alpha\beta)^{m-1/2}}{(m-1)!m!}. \quad (30)$$

Its last Gamma quotient tends to $\Gamma(2)/\Gamma(m+1) = 1/m!$; the other fixed factor is $\Gamma(m) = (m-1)!$. Dividing the coefficients proves (10).

6. An absolutely separable constant-fraction subset

Let $\mathcal{K} = \{\lambda_{\max} \leq (\beta/\alpha)\lambda_{\min}\}$. By [KBNKB, Supplemental Lemma 2] every state with such a spectrum is separable. The criterion is spectral, so it implies absolute separability and $\mathcal{K} \subseteq \text{AS} \subseteq \text{APPT}$. This is an external established criterion, not a new assumption about $\text{AS} = \text{APPT}$.

Normalize independent mean-one exponential variables E_i by their sum; the result is uniform on the probability simplex. The Jacobian of $E_i = t\lambda_i$ is t^{D-1} . The ratio event is homogeneous. Integrating over the unique minimum gives the exact volume

$$\begin{aligned} V_K &= D \int_0^\infty e^{-Dx}(1 - e^{-x/\alpha})^{D-1} dx \\ &= \alpha DB(\alpha D, D) = \frac{\Gamma(\alpha D + 1)\Gamma(D)}{\Gamma(\beta D)}. \end{aligned} \quad (31)$$

Use $u = e^{-x/\alpha}$ for the beta integral. Stirling gives

$$V_K = \sqrt{2\pi\alpha\beta}\sqrt{D}q_m^D(1 + O_m(D^{-1})). \quad (32)$$

This lower bound and Theorem 1 prove (6). More precisely the lower and upper limits of $V_{\text{AS}}/(\sqrt{D}q_m^D)$ lie between $\sqrt{2\pi\alpha\beta}$ and C_m . The exact leading AS coefficient for $m \geq 3$ remains undetermined. In particular AS occupies a fraction bounded away from zero of T for each fixed m .

7. Conditional spectral limit shape

Under T , equations (17) and (19) give

$$D\lambda_i = 1 + \sum_{j < D} d_{ij}y_j, \quad d_{ij} = c_j[(D/j)\mathbf{1}_{j \geq i} - 1]. \quad (33)$$

Uniformly in i, j , $|d_{ij}| \leq 2m$ for all sufficiently large D . If $j < i$ the bound follows from $c_j \leq 2m$. If $j \geq i$, the three ranges in (19) give upper bounds $m/j \leq m$, $(D - j)/(j + \alpha D) \leq 1/\alpha \leq 2$, and $m/(D - m)$, respectively. Thus (24) yields

$$\text{Var}_T(D\lambda_i) \leq 4m^2/(D + 1). \quad (34)$$

The mean is

$$\mathbb{E}_T(D\lambda_i) = 1 - D^{-1} \sum_{j < D} c_j + \sum_{j=i}^{D-1} c_j/j. \quad (35)$$

Away from finitely many boundary indices, $c_j = j/(j + \alpha D)$ and $c_j/j = 1/(j + \alpha D)$. Replacing boundary terms by these formulas changes (35) by $O_m(1/D)$, uniformly in i : the first sum has a factor $1/D$, and every boundary term in the second sum is itself $O_m(1/D)$. Smooth Riemann-sum estimates therefore show uniformly that the mean differs by $O_m(1/D)$ from

$$1 - \int_0^1 \frac{s}{s + \alpha} ds + \int_{i/D}^1 \frac{ds}{s + \alpha} = \alpha \log(\beta/\alpha) + \log \frac{\beta}{\alpha + i/D}. \quad (36)$$

At a fixed finite grid of indices, (34)–(36) and Chebyshev imply convergence in probability. All spectra in T are ordered, even when their least coordinate is negative. Monotonicity between grid points and uniform continuity of Q_m give convergence uniformly over all indices: choose the grid for a fixed tolerance, then let D grow. The two endpoint indices are 1 and D and converge to $Q_m(0)$ and $Q_m(1)$. This proves (7) under T .

For APPT, its T -probability tends to one by (5). For AS, (32) and (22) give a positive lower limit for its T -probability. For any bad event B , $\Pr_T(B \mid E) \leq \Pr_T(B)/\Pr_T(E)$, so the same uniform convergence holds after conditioning on either property. Equality of those two sets or their volumes is not needed.

The monotone change of variables $s = \beta e^{a-x} - \alpha$ shows that $Q_m(U)$ for U uniform on $[0, 1]$ has density (8). Its integral is $\beta - \alpha = 1$, using $e^{a-b} = \alpha/\beta$. Its first moment is $\beta(a + 1) - \alpha(b + 1) = 1$, and its second moment is

$$\beta(a^2 + 2a + 2) - \alpha(b^2 + 2b + 2) = 2 - \alpha\beta L^2. \quad (37)$$

Uniform convergence (7) and Riemann summation imply convergence of the empirical measure and the second moment in probability. Endpoints and the ratio follow because $a > 0$. This completes Theorem 2.

8. Verification boundary and remaining problems

The ancillary checker verifies rational identities, slot counts, simplex moments, Gamma-product identities in finite regressions, leading-coefficient specializations, symbolic density moments, and deliberately corrupted controls. It is not a finite-verification substitute for the all-dimension analytic proof above. This result has not been formalized in Lean. The earlier frozen qutrit purity Lean proof is a different result and does not certify this note.

Unsuccessful three-level numerical purity searches motivated the switch to this complete geometric law. They are not premises. The unrestricted maximal-purity conjecture, the exact AS leading coefficient in $m \geq 3$, proportional growth of both local dimensions, and non-flat matrix measures remain distinct unsolved targets here. AI assistance is disclosed; no external peer review or exhaustive priority verification is claimed.

References

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Disclosure and availability

This research was conducted independently and received no external funding. AI assistance was used for research exploration, analytic derivation, exact-checker development, and manuscript preparation. The claims are presented for independent mathematical scrutiny; no external peer review is represented. The preceding qutrit Lean theorem is distinct from the present analytic proof. Sources, exact ancillary checks, deliberate-error controls, and verification records are available in the public repository `mxym/math`, under `research/appt-spectral-volume-asymptotics/`. No new blanket license or arXiv submission is asserted.